Singapore School Kinderland
Secondary-1
Computer Science
(Chapter 1: Sections 1.1 to 1.1.3)
Term 1

Class Test-1 (SECTIONS 1.1 TO 1.1.3 — ANSWER KEY & MARK SCHEME)

Date: _____ **Name: _____**

Total marks: 15 **Marks obtained: _____**

Teacher's signature: _____ **Parent's signature: _____**

Question 1: The Binary Number System & Denary Conversion [Total: 4 marks]

Textbook Reference: Section 1.1.1 & 1.1.2 (Pages 2–7)

(a) Computers only process binary data (0s and 1s). Explain why computers use the binary number system to represent and process data. **[1 mark]**

Answer / Mark Scheme:
Computers consist of electronic circuits, logic gates, and transistors that operate using electrical signals with only two stable physical states: ON (voltage present / 1) and OFF (no voltage / 0) [1 mark].

(b) Convert the following 8-bit binary number into a denary number. Show your working:

0 1 1 0 1 0 1 1 **[1 mark]**

Answer / Mark Scheme:
Place values: 64 + 32 + 8 + 2 + 1 = 107
Denary: 107 [1 mark]

(c) Convert the denary number 174 into an 8-bit binary number using successive division by 2 or place-value subtraction. Show all working steps clearly. **[2 marks]**

Answer / Mark Scheme:
Method 1 (Successive division by 2):
174 / 2 = 87 r 0; 87 / 2 = 43 r 1; 43 / 2 = 21 r 1; 21 / 2 = 10 r 1;
10 / 2 = 5 r 0; 5 / 2 = 2 r 1; 2 / 2 = 1 r 0; 1 / 2 = 0 r 1

Reading remainders upwards / reverse: 10101110 [1 mark for working]
8-bit Binary: 10101110 [1 mark for final answer]

Question 2: Hexadecimal Number Conversions [Total: 4 marks]

Textbook Reference: Section 1.1.2 (Pages 8–12)

(a) Convert the 12-bit binary number 0111 1101 0100 into Hexadecimal notation. Show your working. **[1 mark]**

Answer / Mark Scheme:
Split into 4-bit nibbles: 0111 | 1101 | 0100
0111 = 7, 1101 = 13 (D in hex), 0100 = 4
Hexadecimal: 7D4 [1 mark]

(b) Convert the hexadecimal value 5B8 into an unsigned binary number. **[1 mark]**

Answer / Mark Scheme:
Expand each hex digit into 4 bits: 5 = 0101, B (11) = 1011, 8 = 1000
Binary: 010110111000 [1 mark]

(c) Convert the hexadecimal number 2C5 into a denary number. Show your full calculation. **[2 marks]**

Answer / Mark Scheme:
Calculation: (2 * 16^2) + (C * 16^1) + (5 * 16^0)
= (2 * 256) + (12 * 16) + (5 * 1)
= 512 + 192 + 5 = 709 [1 mark for calculation steps, 1 mark for correct denary value: 709]

Question 3: Denary to Hexadecimal Conversion [Total: 3 marks]

Textbook Reference: Section 1.1.2 (Pages 11–12)

(a) Convert the denary number 490 into Hexadecimal. Show your successive division by 16 method and calculations clearly. **[2 marks]**

Answer / Mark Scheme:
Working:
490 / 16 = 30 with remainder 10 (A in hex)
30 / 16 = 1 with remainder 14 (E in hex)
1 / 16 = 0 with remainder 1
Reading remainders upwards / reverse: 1EA [1 mark for division/method]
Hexadecimal: 1EA [1 mark for correct hex value]

(b) State how many hexadecimal digits are needed to represent the denary value 490. **[1 mark]**

Answer / Mark Scheme:
3 hexadecimal digits (1, E, A) [1 mark]

Question 4: Uses of Hexadecimal & HTML Colour Codes [Total: 4 marks]

Textbook Reference: Section 1.1.3 (Pages 13–15)

(a) State two reasons why software developers and engineers prefer to use hexadecimal notation instead of pure binary notation. **[2 marks]**

Answer / Mark Scheme:

1. Hexadecimal is much more compact / takes up significantly less screen space (1 hex digit = 4 binary bits) [1 mark].
2. It is much easier for humans to read, write, and remember, reducing transcription and debugging errors [1 mark].

(b) (i) In web design, colours are defined using 6-character hexadecimal codes in HTML (#RRGGBB). State which colour is represented by the code #FF0000. **[1 mark]**

Answer / Mark Scheme:

Red (Full intensity for Red, zero for Green and Blue) [1 mark]

(b) (ii) Explain what the hexadecimal colour codes #000000 and #FFFFFF represent in terms of RGB light intensity. **[1 mark]**

Answer / Mark Scheme:

#000000 represents Black (all RGB color channels at minimum/zero intensity: 0, 0, 0).
#FFFFFF represents White (all RGB color channels at maximum intensity: 255, 255, 255) [1 mark].